

Intelligent choreography movement sequence generation and optimization based on deep reinforcement learning

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ABSTRACT

Choreography for traditional dance is mostly human-driven, based on creativity and artistic intuition, making it difficult to automatically generate and explore novel dance movements in large scale. This challenge is particularly important for tasks like creating expressive and realistic dances from motion capture and interactive motion control in systems like motion capture, virtual reality, and entertainment systems. It is not feasible to present long-range temporal dependencies and continuous motion dynamics using existing rule-based approaches and traditional machine learning. In order to solve this issue, a deep reinforcement learning-based choreography generation framework, called choreography generation framework based on deep reinforcement learning (CMSG-DRL), is proposed that is designed as a Markov Decision Process (MDP). A policy network based on a transformer is used to model long-term temporal dependencies in dance sequences and a secondary validation module is used to ensure physical plausibility and motion feasibility. A multi-objective reward function is created to optimize the aesthetics, biomechanical correctness and music synchronization simultaneously to generate balanced choreography. To promote exploration and stabilize motion learning, the Soft Actor-Critic (SAC) algorithm with entropy regularization is used. The model is trained on a high-quality motion capture dataset of expert dancer, which has a wide variety of motion patterns and is realistic. The experimental results show that the proposed CMSG-DRL improves the accuracy of the timing with an average of 15.2%, the smoothness of the movement by 14.8%, and the creative quality of the movement by 15.6% over the best known approach, which proves its efficiency to produce expressive and physically consistent motion.

1. Introduction

Computational creativity is a rapidly developing field in artificial intelligence that addresses systems undertaking creative tasks, such as painting, composing music, or creating dance [1]. Creativity technologies are transforming the way artists create, rehearse, and experiment with ideas in the performing arts [2]. Dance is becoming a priority area since it is intricate and susceptible to rich interaction with machine learning methodology [3]. Systems that learn to produce movement dynamics and style can assist human choreographers and extend the horizons of conventional choreography [4]. The discipline brings together movement science, machine learning and art theory to build a next generation intelligent system that can model and generate expressive motion [5]. These systems can take away tedious mechanical work from choreographer and maintain the emotional and physical authenticity of dance [6]. The space, rhythm, balance and beauty-perceiving algorithms can help choreographers find new patterns or automatically perform elements of their work that are

repetitive. These kinds of tools are being utilized in professional productions and educational institutions to create or propose movement patterns based on data and style inputs [7].

Human motion generation in different areas, such as animation, robotics, video games and video humans, has been most promising for models of artificial intelligence [8]. These models are designed to reproduce realistic body dynamics and time evolution, to produce smooth and physical relevant motion transitions [9]. Recurrent neural networks, GANs, and reinforcement learning have allowed machines to learn complex motion patterns and have long-term dependencies from huge databases [10]. These developments have become the backbone to AI-generated choreography systems. In choreography generation, these technologies can be used for directly learning movement patterns from motion capturing data, and synthesizing realistic dance sequences [11]. The algorithms for machine learning now produce results that are consistent in style in the presence of body constraints and music rhythm. However, the balance between creativity and realism still exists [12]. Reinforcement learning, especially when the action space is continuous,

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offers a promising answer to these problems, as agents can learn to solve a broad spectrum of problems and to achieve multiple goals. It [13] allows the policy based learning along with the designed reward.

1.1. Problem statement

Planning choreography is typically a human, time-consuming process, and is dependent on the creative insight and expertise of a choreographer. This limits diversity and creativity of movement patterns, leading to dull or stylised performances. The computer techniques used today, like the rule-based one or the model of sequence prediction, are generally very rigid and fail to generate expressive and physically natural dance movements that meet aesthetic and biomechanical guidelines. Also, the models are not suitable for fitting musical input or real-time synchrony with rhythm.

The challenge is to develop an innovative system which could generate choreographic ideas that are innovative, believable and well timed to the music. The system needs to deal with continuous motion data, understand the temporal dependencies and achieve multiple objectives simultaneously. The traditional supervised learning is not sufficient in this case as it is static and does not give adaptive feedback. Therefore a learning model that can not only create movement possibilities but also narrow them down with the assessment of performance needs to be designed. To meet this challenge, this paper proposes to use deep reinforcement learning using a multi-objective reward system.

1.2. Research methodology

In order to overcome these drawbacks, the optimal choreography policy is learned by interacting with the environment, and it can be used to generate expressive dance movements adaptively and with data-driven strategies without relying on a set of rules. The CMSG-DRL system designs choreography movement generation as a continuous control task from the perspective of deep reinforcement learning. The temporal dynamics are learned directly and applied to generate movement sequences that represent both style and structure using a transformer-based policy network. A critic network critiques each sequence based on a multi-objective reward function that considers aesthetic value, physical viability, and musical synchronization. Training is performed using the SAC algorithm to ensure stability and practical exploration in high-dimensional action spaces. The entropy regularization promotes varied movement generation with minimal repetitiveness. Training is conducted using motion capture data from expert dancers, allowing the model to learn realistic motion patterns. CMSG-DRL aims to optimize flow, creativity, and musicality for dance sequences.

1.3. Contributions of the paper

- ❖ To build an intelligent choreography generation framework based on a CMSG-DRL to generate expressive, coherent and music-synchronized transitions of dance pieces with deep reinforcement learning.
- ❖ To develop a policy network that can learn temporal dynamics and complex sequential relationships in dance motion sequences by using a transformer.
- ❖ To define a multi-objective reward function to balance aesthetic quality, biomechanical feasibility, rhythmic synchronization and temporal smoothness of produced movements.
- ❖ To implement the Soft Actor-Critic (SAC) algorithm with the addition of entropy regularization that is able to learn a stable policy and to encourage diversity and creativity in generated choreography.
- ❖ To use and process a professional well-captured dataset (AIST++) to produce and assess a more realistic and physically plausible dance movement.

The key difference of music-conditioned dance choreography

generation with the standard CMSG-DRL framework is the combination of deep reinforcement learning and music-conditioned dance choreography generation. The proposed method contrasts with supervised learning-based methods or generative models like LSTM, Gated Recurrent Unit (GRU) and Generative Adversarial Network (GAN) that train a network to generate choreography in a supervised manner. The proposed method differs from supervised learning-based methods, and generative models like LSTM, Gated Recurrent Unit (GRU) and Generative Adversarial Network (GAN) that learn to generate choreography sequence in a supervised way. This change enables the system to learn choreography in an interactive, as opposed to a sequential, control problem.

An illustration to prove the effectiveness of the proposed framework in practice, one can consider a sample music sequence with a tempo of 120 BPM and specific rhythmic beat patterns. Based on this input, the system removes time-related features like beat positions and rhythm strength that help to create synchronized movement of dances. As an example, when the rhythmic part is strong, the resulting motion can be strong steps or direction changes, whereas when the rhythmic part is weak, the resulting movement can be gentle steps or direction changes. These features are processed in a sequence by the model and generate corresponding 3D pose vectors per frame which are temporally synchronized with the music. This demonstrates how the proposed method can create coherent, rhythm-aware choreography, based on the structured audio input.

One of the drawbacks of rule-based choreography systems is that they are rigid due to the inability to account for the different musical structures and dancer-specific differences by manually defining the motion rules and transition constraints. These systems are based on pre-established input-output pairings, restricting their capacity to generalize to various styles of dance, rhythms, and motion patterns that they have not seen. Therefore, any changes in rhythm or in the dynamics of the poses (even small ones) must first be configured manually, and the system is not suitable for complex choreography generation problems or for generating choreography in real time. Data-driven and reinforcement learning methods, on the other hand, can learn distribution functions of movements from data, allowing for more adaptability and expressive variability in danced sequences.

1.4. Structure of the paper

The rest of the paper is organized in this manner. Section 2 introduces related works in motion generation and dance synthesis. Section 3 describes the introduced deep reinforcement learning model, architecture, and training scheme. Section 4 describes the experiment setup and dataset. Section 5 describes the results and comparison. Section 6 summarizes the research and suggests directions for future AI-assisted choreography generation.

2. Related works

Choreography synthesis using machine learning is currently a multi-disciplinary field of research that spans computer science, human motion analysis and digital performance art. With the advancement of machine learning and motion capture technology, researchers have suggested various systems that use these technologies to create realistic and expressive dance sequences that can react to environmental signals, such as music. This section briefly reviews previous investigations and recent progress of the research areas that are relevant to the CMSG-DRL system.

Ma et al [14]. proposed a computer choreography data-driven system using 3D technology and Kinect. It was developed to collect as little movement data as possible, and to provide the most accurate production of automated dance choreography. The system could generate choreographable movements which quantitatively matched the real human poses in 3D space to perform more realistic virtual performance. The

method was successful in controlled situations and did not perform well in dynamic scenarios and fine movements caused by hardware limitations and issues with fine movements from the Kinect sensor.

Hernandez-Resendiz et al [15]. proposed a data-driven approach, called Process Choreography Discovery (PCD), to discover structured workflows from event logs. It was proposed as a way to improve the automation and intelligency of the business process models by intelligent choreography discovery. The technique was shown to be very good for identifying patterns and sequence relations between the various primary activities with a high degree of accuracy in a corporate context. However, it was not suitable for artistic or real-time movement generation usage such as dance choreography and therefore was not as effective for human or creative movement generation applications.

In order to help the choreography design, Gao [16] created a Computer Three-Dimensional Auxiliary System (CTDAS). It was designed for the benefit of choreographers to create and refine movement sequences through 3D modelling and simulation. The results showed the system contributed to the improvement of accuracy in dance planning and improved spatial awareness of body posture and direction. The method, however, did not possess generative capacity, and was dependent on manual input from the choreographer, thus not being able to produce new or stylistically different choreography on its own.

Au et al [17]. proposed the graph-structured music-conditioned dynamic automatic dance choreography model called "Choreograph". It is added to keep tempo and style when creating dance movements from music inputs. The model produced dances with a high correlation with music's rhythm and stylistic cohesion regardless of genre. While the system was successful in reproducing the rhythms to a similar degree, there was no significant generalization across unseen music from the other genres, and the system was very dependent on the amount of labeled training data.

Zhang [18] developed a deep learning approach to classify variations in folk dance movements based on temporal and spatial features. The intention was to preserve cultural variations of dance and provide movement classification to digital repositories. The model generated high classification accuracy for regional style and movement pattern classification. The model was limited in its analysis and incapable of generating movements or sequences, rendering it inadequate for choreography design or sequence generation.

Eum et al [19]. also suggested a method for dance sequence generation that enhances coherency introduced by Recurrent Convolution Blocks (RCB) technique, named Dance Recalibration using Recurrent Convolution Blocks (DR-RCB). It tried to minimize drastic changes in movement and enhance temporal flow. The method successfully produced smoother dance sequences and on motion continuity assessment was more effective than baseline models. However, it did not rely on external stimuli like music or sense of aesthetics and this affected its ability to perform in rhythm sensing and artistic choreography tasks.

Mao et al [20]. proposed a human motion generation model that is generic and can be generalized using reinforcement learning. The proposed model was intended to create a variety and realistic motion for tasks and environment. The model demonstrated outstanding generalization ability and motion synthesis flexibility. However, the method was designed for ordinary human movements and not for sequences that are stylistic or music related; therefore, it does not fit in the application of dance choreography, which considers the stylistic and rhythmic aspect.

Petrovich et al [21]. proposed a Multi-Track Timeline Control framework to generate 3D human motion from text. Fine-grained control of the movement sequence was proposed, in the form of text inputs. The system proved to be effective for generating a context-aware motion from natural language input for interacting with multiple characters. It was not precise in its control, since it didn't focus on the different aspects relevant to dance, such as beat alignment or aesthetic structure, and so was not suitable for choreography generation tasks.

Chao and Yingren [22] proposed a dance action planning method for

Table 1

Comparative Analysis of Existing Choreography and Motion Generation Techniques.

Ref.	Technique / Model	Advantages	Limitations
[14]	Kinect-based choreography system	Realistic 3D pose approximation; automated choreography generation	Limited motion fidelity; poor dynamic transitions
[15]	Process Choreography Discovery (PCD)	High accuracy in workflow pattern extraction	Not suitable for artistic or real-time motion
[16]	CTDAS (3D choreography support system)	Improves spatial visualization and planning	No generative capability; manual input required
[17]	Choreograph (graph-based model)	Strong rhythm synchronization; style consistency	Poor generalization; requires large labeled data
[18]	Deep learning-based dance classification	High classification accuracy; preserves cultural styles	No sequence generation capability
[19]	DR-RCB model	Smooth motion generation; improved temporal coherence	No music or aesthetic integration
[20]	RL-based motion generation	Strong generalization; diverse motion synthesis	Not tailored for music-driven choreography
[21]	Text-driven motion generation	Context-aware motion; multi-character interaction	Lacks beat alignment and choreography structure
[22]	Robot dance planning system	Simple interactive dance generation	Template-based; lacks adaptability
[23]	MEGADance (Mixture-of-Experts)	High style fidelity; genre-conditioned synthesis	Requires large dataset; poor rare-style handling
[24]	Beat-aligned motion analysis	Accurate beat detection and segmentation	No generative capability
[25]	Dance aesthetics study	Improves understanding of visual perception	Not applicable for motion generation
[26]	Abductive knowledge reasoning	Strong semantic alignment; robust inference	High computational cost; scalability issues
[27]	Relational semantics augmentation	Improved alignment accuracy; better generalization	Sensitive to noise; increased model complexity

entertainment robots, which was created by using a machine learning based teaching game mode. It was proposed to support interactive educational of entertainment robotics and art education. The system worked well to plan and execute basic dancing movements according to external control. However, the approach was based on pre-specified actions templates and did not involve adaptive learning and sequence optimization, which limited its applicability when it comes to advanced autonomous choreography systems.

Yang et al [23]. introduced MEGADance, a type of Mixture-of-Experts architecture for 3D dance synthesis conditioned by genre. MEGADance was developed to maintain rich dance styles and produce genre-conditioned sequences from musical inputs. MEGADance attained excellent style fidelity and rhythm compliance for various genres. However, the model needed large quantities of genre-tagged data and performed extremely poorly when handling rare or underrepresented styles, thereby constraining its scalability and real-time responsiveness for freestyle or hybrid dance synthesis.

Shen & Hirayama [24] proposed a beat-aligned motor synergy detection and street dance kinematic beat tracking method. The use of analysis to synchronize body parts with beats and assist with movement classification was recommended. The model was able to perform efficient beat detection and movement segmentation, and help with the analysis of dance behavior. The system was not analysis-based, but lacked the generative ability to create new or creative dance sequences.

Kempe & Heinen [25] investigated the effect of stage arrangements on the appreciation of the aesthetic value of dance performances. It was suggested to understand the visual impact on audience ratings and

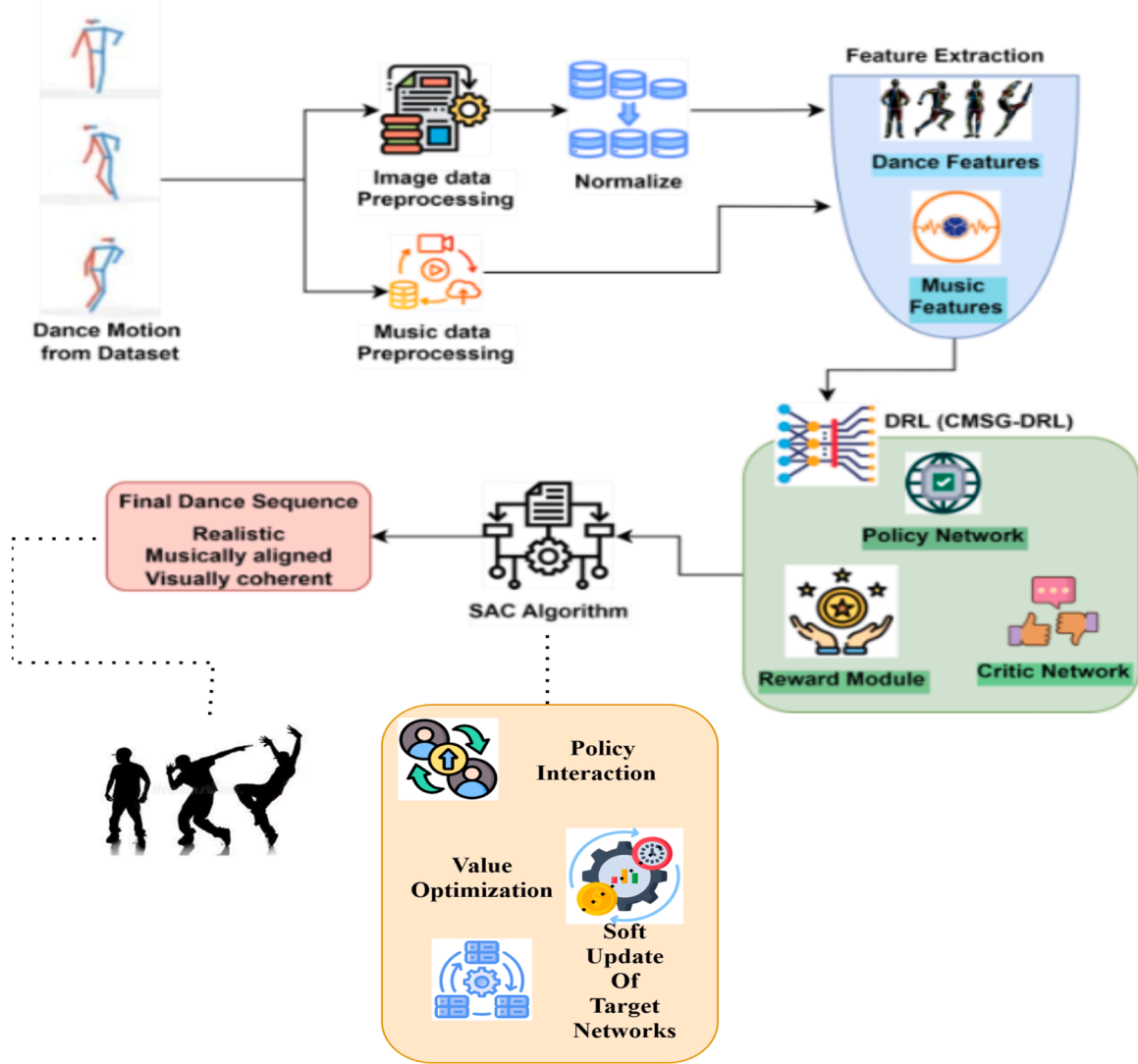


Fig. 1. Architecture of the CMSG-DRL Framework.

enhance the choreography staging. The findings indicated that space layout, setting, and lighting had a significant effect on perceived attractiveness and interest in the performance. The generation of movement or intelligent choreography is apparent from the research, which is confined to automated or AI-based dance systems.

Akhtar et al [26]. presented an abductive knowledge reasoning across multiple knowledge graphs based multilingual entity alignment method. The approach combines semantic embeddings with logical reasoning to identify relationships between entities from different languages, resulting in improved alignment accuracy and robustness. The method, however, requires high quality knowledge graph representations and it has a higher complexity. Akhtar et al [27]. proposed a relational semantics augmentation framework to improve the entity representation by structural and semantic relationships of the multilingual knowledge graph. By using relational context they were able to generalize more and perform better in the alignment task, but their method is sensitive to noisy data and requires careful parameter tuning. Both studies show the benefits of semantic reasoning and semantic representation learning; however, they are not temporal motion generation or choreography synthesis methods, emphasizing the necessity of developing specific models like the proposed CMSG-DRL model.

Table 1 provide comparative analysis of the currently used choreography and motion generation techniques according to their main

advantages and disadvantages. It also shows the strength of different methods in regards to realism of motion, synchronization, and adaptability to learning. The comparison also highlights the critical research gaps, which creates a need to have a high-level framework like the proposed CMSG-DRL model.

Online fault diagnosis of industrial robots based on an IoRT framework of hybrid deep learning was proposed by Bilal et al [28]. This study was aimed at the problem of not only detecting faults in real-time industrial environments but also detecting them at the right time and in the correct position. The proposed approach showed a high level of diagnostic accuracy and improved system reliability, but was very sensitive to the quality of the sensor data and its accuracy decreased with noisy IoT conditions.

Kamal et al [29]. proposed a deep reinforcement learning (DRL) method for adaptive traffic signal control (ATS) using a digital twin. This work centered around addressing the issue of static traffic management systems which are unable to adjust to dynamic congestion. The proposed method has potential to enhance traffic flow efficiency and decrease waiting time, however it requires highly computational resources and accurate digital twin modeling.

Zheng et al [30]. proposed a split federated learning framework for vehicular semantic communication networks that accounts for vehicle mobility. The study discussed the data heterogeneity and

communication inefficiency in the vehicular systems. The suggested method enhanced learning effectiveness and communication performance, but it is vulnerable to learning errors in mobility predictions and to network instability.

To address the spatiotemporal correlation in data collection in an energy-constrained UAV-assisted IoT system, Xu et al [31]. proposed a UAV-assisted data collection framework. The work was focused on problems of energy inefficiency as well as on large-scale data gathering in IoT networks. The method enhanced energy and data-collection efficiency but could not be scaled up in dense IoT environments and relied on the UAV's operational stability.

Ma et al [32]. introduced an intelligent engineering systems optimization framework based on deep learning. The study aimed to solve the problems of traditional optimization algorithms in complex engineering environments. The proposed model was able to enhance the accuracy of the optimization and increase the efficiency of the system, but it needed large data sets and significant computer resources.

Ali et al [33]. suggested more sophisticated machine learning methods to facilitate adaptive optimization of complex systems. The study was aimed at addressing the shortcomings of the traditional methods in the nonlinear and dynamic systems. The proposed approach is more adaptable, converging faster but with low interpretability and high complexity of parameter tuning.

Müller et al [34]. proposed a hybrid deep learning approach to intelligent predictive modeling. The study focused on the issues of dealing with uncertainty and non-linear temporal dependency in prediction tasks. The proposed model was found to be more accurate and robust, but the hyperparameter optimization is sensitive to the selection of hyperparameters and needs large amounts of training data.

Iraji [35] presents a semi-supervised GAN model for skin lesion diagnosis with imbalanced data. The model is based on an unbiased generator and utilizes both labeled and unlabeled data to enhance feature learning and synthesize informative data, especially for minority classes. The main challenge tackled is class imbalance in the datasets of skin lesions, causing overfitting in traditional deep learning models and limiting diagnostic accuracy. The experimental results show that the proposed method enhances the classification accuracy and provides more balanced accuracy across all the classes as compared to the existing supervised and baseline GAN based classification approaches. The study has several drawbacks, such as the high complexity of GAN training, possible difficulties in deployment in real-time settings, and a need for additional verification on a variety of clinical data sets to achieve better generalisability.

The field of intelligent systems and optimization offers foundations to learn based motion and control frameworks. The research field of intelligent systems and optimization offers solid ground for learning based motion and control. Progressive iterative approximation methods have been used for compressing video data, which are high dimensional temporal sequences, in an efficient manner maintaining the structural information. The modified secant-based BFGS approaches enhance convergence and stability for unconstrained learning problems in optimization.

Robots with three degrees of freedom are used in rehabilitation to aid in the recovery of human movement in an adaptive manner, by using precise control strategies, in the case of a knee and ankle rehabilitation robot. In warehouse environments, human-robot communication systems based on gesture based interaction improve human-autonomous agent coordination. Furthermore, soft trunk robot research concerns modeling dynamics of motion and fault detection for enhancing the stability of deformable robotic structures.

3. CMSG-DRL methodology

This section introduces the architecture and operating paradigm of the target CMSG-DRL system-Choreography Movement Synthesis and Generation with Deep Reinforcement Learning. The key goal is to

Table 2

Summary of the pre-processing steps.

Preprocessing Steps	Description	Equation
Joint Position Extraction	Represents each frame as a 3D skeletal pose vector across J joints	$M = \{p_1, p_2, \dots, p_T\}, p_t \in \mathbb{R}^{3J}$ $p_t = [x_1, y_1, z_1, \dots, x_J, y_J, z_J]$
Music-Beat Synchronization	i. Extracts frame indices corresponding to beat-aligned musical events. ii. Segments motion sequences aligned with each beat.	$B = \{b_1, b_2, \dots, b_K\}, b_k \in N$ $S_k = \{p_{b_k}, p_{b_k+1}, \dots, p_{b_k+\Delta}\}$ $p_t^{norm} = p_t - r_t$
Translation Normalization	Converts poses to root-relative coordinates by subtracting the root joint position.	$p_t^{aligned} = R_t p_t^{norm}$
Rotation Normalization	Aligns character heading to a canonical forward direction using yaw rotation.	
Scaling Normalization	Normalizes body size using average limb length to ensure invariance to dancer height.	$p_t^{final} = \frac{p_t^{aligned}}{l_{avg}}$
Fixed-Length Segmentation	Extracts overlapping motion windows of fixed length	$S_i = \{p_i, p_{i+1}, \dots, p_{i+L-1}\}$
Music Feature Pairing	Pair each motion segment with the corresponding tempo and rhythmic structure.	$D = \{(S_i, m_i)\}_{i=1}^N$

synthesize natural-sounding and music-synchronized dance movements by learning challenging-to-model relationships between motion dynamics and audio features. The research is organized around an MDP, enabling the agent to learn optimal motion policies in interaction with an environment that provides rewards or penalties. The system input consists of synchronized 3D dance motion data and corresponding music features, which are used to derive informative spatiotemporal embeddings through processing. They are used to specify the state representations in a reinforcement learning setting, where an SAC agent is learning to generate dance sequences, step by step. Key elements of the framework include pre-processing of the data, feature extraction, formulation of the MDP, policy learning and choreography generation. The actions needed for each component are described in turn in the subsequent subsections. Figure 1 shows the architecture of the CMSG-DRL framework.

3.1. Data collection and preprocessing

The training sequence of choreography is a 3D motion capture sequence with high resolution given by AIST++ data set. It includes “professional dance movements” at 60 frames per second along with “music audio” that is synchronised to the dance in real time. It captures the full-body skeletal joint positions in the 3D space and is a continuous series of dance movements over time. The music recordings that come with an associated device are processed using conventional audio signal processing to extract the beat-aligned features such as rhythm, tempo and locations of the beat. These properties are used to create a temporal relationship between the movements and music, and are integral to the production of musically uniform choreography.

3.2. Preprocessing motion and music data

To pre-process the raw motion and music data for training, a formal preprocessing pipeline is used. The 3D positions of the joints are first obtained for every frame to create sequential pose vectors. Pose sequences are then aligned with the music's beat markers so that dance steps are synchronized with the rhythm. To remove inter-dancer variability and normalize motion representation, the joint positions are translated (root-relative), rotated (heading alignment), and scaled (body-size invariance) to normalize. Table 2 summarizes the pre-processing steps. Sequences are segmented into overlapping fixed-length

windows following normalization. Each window exhibits a consistent movement pattern that maps to a corresponding musical window. Paired sequences and beat-aligned music features are used as input to train a model to generate choreography.

In the preprocessing stage, the key factor in achieving stable learning without sacrificing cross-genre diversity in dance is normalization. Translation-normalization transforms the joint positions into root-relative coordinates, which eliminates the positional bias of the joints from the global coordinate system but still preserves the relative movement patterns between the joints that are important for style learning. Rotation normalization aligns the dancer's orientation with the canonical forward direction, minimizing the effect of viewpoint while preserving intra-body articulation dynamics, enabling consistent learning of movement across a wide range of dance styles.

The scaling normalization process allows for body-size invariance by normalizing the lengths of all limbs, avoiding a dancer body-size bias. Stylistic diversity is not reduced in this transformation because the features of different genres are not coded in absolute space but in temporal motion dynamics and in features of music conditioned by those dynamics. where $p_t \in \mathbb{R}^{3J}$: Frame t 's position vector with 3D coordinates (x_J, y_J, z_J) of each one of the J joints, r_t = Global 3D space position vector of root joint (usually pelvis) for translation normalization in frame t , R_t = Rotation matrix calculated to align the orientation of the skeleton to a known forward direction so that it is normalized for all dancers. l_{avg} = The average of the lengths of limbs over the skeleton, to position all body sizes on a standard scale. S_k = A pose segment beginning at a beat-synchronized frame b_k and of length Δ frames. L : The length (in frames) of the fixed-size overlapping window used for segmentation during training. m_i : A vector of music features (tempo, rhythm, beat timing) aligned with the segment S_i . $D = \{(S_i, m_i)\}_{i=1}^N$ = The final training dataset is composed of NNN paired motion-music segments.

3.2.1. Token reduction module

The graph-conditioned token selector exploits the input sequence by constructing a motion graph in which each pose token is a node and each spatiotemporal dependency is an edge. The relevance scores are calculated using graph attention (pose dynamics and edge continuity), and high-scoring tokens are retained, while low-scoring redundant tokens are discarded.

For example, key motion transitions that involve jump and gesture aligned beats score higher ($\approx 0.82, 0.79, 0.76$). They are retained, which preserves the essential structures of the choreography, but also cuts down on the choreography's repetitive poses and removes poses with low variation ($\approx 0.21, 0.18$).

3.3. Feature extraction

To ensure successful learning in the reinforcement learning (RL) environment, features from dance and music are extracted, represented, and even time-aligned. They are the state representations used by the proposed CMSG-DRL model to generate choreography sequences. The extracted features convey the sense of body movement, rhythm, and musical architecture, creating dance that is both physically and musically aligned.

This study has implemented the feature extraction process in the open-source repository, making it fully reproducible. The feature engineering module preprocesses raw input data into a structured format suitable for model training, including normalization and other domain-specific transformations. The implementation scripts are included in the repository for transparency and reproducibility of the preprocessing pipeline. This is the evaluation dataset, publicly available via the AIST++ dataset visualizer platform: https://google.github.io/aistplusplus_dataset/visualizer/index.html?c=gHO. Large-scale annotated motion data is introduced in [36] for research and benchmarking of motion analysis tasks.

Table 3

Summary of the dance feature extraction steps.

Dance Feature Extraction Steps	Equation	Description
Pose Embedding	$e_t = W_p \cdot p_t^{final} + b_p$	Projects 3D pose into a lower-dimensional embedding space.
Joint Velocity	$v_t = p_t^{final} - p_{t-1}^{final}$	Captures first-order motion (rate of position change).
Joint Acceleration	$a_t = v_t - v_{t-1}$	Represents second-order motion (rate of velocity change).
Dance State Vector	$s_t^{dance} = [e_t, v_t, a_t]$	Concatenates pose, velocity, and acceleration into a single feature vector.

3.3.1. Dance feature extraction

Dance synthesis is based on the useful features extracted from the sequences of 3D poses, which are normalized. They represent the space form and the time motion of the human body. Pose embeddings are generated by mapping the 3D skeletal pose to a lower dimensional space, or latent space, with a learnable linear operation. This removes duplication, but maintains significant spatial relationships. Second, velocities of joints are computed as the 1st-order differences of pose vectors between frames. This means the velocity and direction of the motion of the joints, thus highlighting the dynamism of the movement of the joints. Third, the second order differences in poses are used to calculate joint accelerations; these represent changes in movement speed. This indicates the strength and magnitude of motion over time. The dance feature extraction process steps are presented in Table 3. The state representation of the dancer at each time step is the combination of these three elements (pose embeddings, velocities and accelerations). This is fed into reinforcement learning for dance.

$p_t^{final} \in \mathbb{R}^{3J}$ = The normalized 3D coordinates of J skeletal joints at frame t . Each joint has three components, x, y, and z. $W_p \in \mathbb{R}^{d \times 3J}$ = Trainable weight matrix used to embed the pose into a latent feature space of dimension d . $b_p \in \mathbb{R}^d$ = Bias vector added after linear projection to embed pose. $e_t \in \mathbb{R}^d$ = Latent pose embedding structural pose information of the body at time t . $v_t \in \mathbb{R}^{3J}$ = Joint velocity vector that captures how fast and in which direction every joint is moving between frames $(t-1)$ and frame (t) , $a_t \in \mathbb{R}^{3J}$ = Joint acceleration vector that captures how the motion dynamics evolve from frame to frame, $s_t^{dance} \in \mathbb{R}^{d+6J}$: Final feature vector capturing the entire kinematic and dynamic state of the dancer at every time step, including pose, velocity, and acceleration.

3.3.2. Music feature extraction

Temporal audio features are extracted from the background music through audio signal processing techniques, mainly with Librosa, in

Table 4

Summary of the music feature extraction steps.

Music Feature Extraction Steps	Equation	Description
Beat Timing	$B = \{b_1, b_2, \dots, b_K\}$	Extract beat-aligned frame indices using onset detection and beat tracking.
Tempo	$\tau = \frac{60}{\text{mean}(\Delta t_{\text{beat}})}$ $\Delta t_{\text{beat}} = b_{(k+1)} - b_k$	Estimate global tempo in beats per minute (BPM).
Rhythm Pattern Encoding	$r_t \in \mathbb{R}^m$	Encodes local beat strength and tempo context from tempogram or STFT-based analysis.
Genre Indicators	$g \in \{0, 1\}^c$	One-hot encoding of musical genre metadata.
Music State Vector	$s_t^{music} = [\tau, r, g]$	Concatenation of rhythm pattern, tempo, and genre features at each time frame.

order to synchronize the movements of the dance. Some of the features that can be extracted are beat positions, tempo and rhythmic structure, and, optionally, some genre metadata. Beat timing is achieved by onset detection and beat tracking algorithms computing the temporal position of detected beats in the audio signal. Perceptual beat indexes are anchors for aligning the motion segments. Tempo represents the average time interval between two successive perceptual beat indications (in beats per minute, or BPM), and is a general indicator of the tempo of a musical. Spectral or periodicity-based analyses of rhythm patterns are used to characterize the local strength and fluctuations in beat timing over short time intervals, such as Short-Time Fourier Transform (STFT) or tempograms. Genre indicators (if available) are coded as categorical features with one-hot vectors. These characteristics enable the adaptation of dance movements towards music genres. The steps of the music feature extraction are shown in Table 4. The full music feature vector for each frame is created by chaining the rhythm pattern vector, tempo and genre metadata together, resulting in an input for the reinforcement learning model which is temporally aligned.

$b_k \in N = \text{Index of the audio frame corresponding to the } k^{\text{th}} \text{ beat in the music signal}$, $B = \{b_1, b_2, \dots, b_K\} = \text{Set of all beat frame indices identified in the audio track}$, $\Delta t_{\text{beat}} = b_{(k+1)} - b_k = \text{Interval (in frames) between successive beats}$, $\tau \in R = \text{Estimated tempo, calculated as the inverse of the mean beat interval, scaled to beats per minute}$, $r_t \in R^m$: Rhythmic feature vector at time t , summarizing local tempo and beat strength over a context window of size m . $g \in \{0, 1\}^c = \text{One-hot vector indicating the genre of the music, where } c \text{ is the number of genres considered}$. $s_t^{\text{music}} \in R^{m+1+c} = \text{Final music feature vector for time step } t$, combining rhythmic and categorical musical information for input to the learning system.

To exploit its improved pretraining strategy, RoBERTa-base is used as the base, which removes the masking of "Dynamic Masking" and makes the Next Sentence Prediction (NSP) objective simpler. This gives rise to better contextual representations which can be more effective for the motion-text alignment task in choreography generation than BERT. Dance specific encoders may be able to learn domain-specific patterns but are not often generalizable because of low number of training samples. Compared with RoBERTa, which has excellent generalization ability in sequential and multimodal tasks, and is more scalable, it is possible to see that it is very similar.

For construction, a random walk method is used to model structural dependence of motion tokens. The walk length will be $L = 6$ and $N = 10$ walks per node, a compromise between the local and global context. This leads to proper neighbourhood coverage of learning relationships of motions and less over-smoothing in the representation of graphs. Overall, it improves the consistency and stability of the choreographic sequence models in structures while keeping the computing efficiency.

3.4. Environmental design (MDP framework)

To model the choreography generation process as a sequential decision-making problem, the system is formulated within the Markov Decision Process (MDP) framework. MDP is characterized as a tuple (S, A, R, T, γ) , with S : States Possible, A : Collection of potential actions, R : Reward function, T : Transition dynamics, and γ : Discount factor (usually $\gamma \in [0, 1]$).

3.4.1. State space ($s_t \in S$)

At time t , the state is the agent's current representation of the choreographic space. It consists of: Current Pose Embedding (e_t) = Low-dimensional body pose embedding in the current time, Motion History $H_t = \{e_{t-h}, \dots, e_t\} = \text{Recent pose embeddings history for the last } h \text{ time steps}$. Music Context (s_t^{music}) = Audio features such as rhythm vector (r_t), tempo (τ), and genre (g). The entire state is represented as in equation 1.

$$s_t = [H_t, s_t^{\text{music}}] \quad (1)$$

In equation 1, $s_t = [H_t, s_t^{\text{music}}]$ represents the construction of the

combined state vector at the time step t . Here, H_t denotes the temporal or contextual feature representation extracted from the current input, while s_t^{music} represents the music-related or modality-specific feature component. The notation $[\cdot, \cdot]$ indicates the concatenation of these feature vectors to form a unified state representation s_t . This fused state captures both temporal dynamics and music-conditioned information, enabling the model to effectively learn correlated multi-source features, improving decision-making and prediction performance.

3.4.2. Action space ($a_t \in A$)

At each timestep, the agent chooses an action that is the next pose vector, p_{t+1} , in the 3D joint space. The action is predicted from the learned latent pose space, taking into account both physical and stylistic constraints. This is shown in equation 2.

$$a_t = p_{(t+1)} \in \mathbb{R}^{3J} \quad (2)$$

Equation (2) defines the action representation in the proposed framework, where $a_t = p_{t+1} \in \mathbb{R}^{3J}$. Here, p_{t+1} predicted pose at next time step, $t+1$ and $\mathbb{R}^{3J}$ represents the 3D coordinate space, which is continuous, of J joints. So, Equation (2) can be put in the form of action vector as the future pose configuration, which contains spatial joint position (x, y, z) for each joint. This formulation allows the model to properly learn the transition between poses over time and to predict sequential poses properly.

3.4.3. Reward function ($R(s_t, a_t)$)

The reward function used to guide the agent to create good dance sequences is a multi-objective reward function. Reward R_t at each timestep is a combination (with weights) of the following: Aesthetic Appeal Score ($R_{\text{aesthetic}}$): Measures the visual smoothness, diversity and stylistic compliance of the learned discriminators or heuristics. Physical Correctness Score (R_{physics}): penalizes unnatural joint angles, unrealistic velocities and Physical Feasibility violations (e.g. foot sliding). Musical Synchronization Accuracy (R_{music}): Effect of synchronization of motion beats and audio beats. Temporal Coherence Score (R_{smooth}): Implies smooth transitions, encourages smooth transitions and penalizes sudden accelerations.

The reward function R_t assigns a weight to each of the components in the CMSG-DRL framework to ensure high-quality dance choreography generation: $R_t = \lambda_1 R_{\text{aesthetic}} + \lambda_2 R_{\text{physics}} + \lambda_3 R_{\text{music}} + \lambda_4 R_{\text{smooth}}$, where each component represents a different quality of choreography generated by the motion. The aesthetic reward $R_{\text{aesthetic}}$ evaluates the aesthetic of the pose and its stylistic integrity, while the physical correctness reward R_{physics} penalizes unnatural joint angles as well as kinematic violations. Reward R_{music} ensures temporal alignment of the motion and musical beats, while penalty $R_{\text{smooth}} = -\|a_t\|_2^2$ reduces abrupt changes in motion by penalizing high acceleration.

The weighting coefficients are: $\lambda_1 = 0.30$, $\lambda_2 = 0.25$, $\lambda_3 = 0.30$, and $\lambda_4 = 0.15$, which balance the artistic level, physical realism, rhythmic alignment and temporal smoothness. This formulation provides the Soft Actor-Critic (SAC) agent with a guidance to create expressive, physically feasible and music-synchronized choreography. The weights were tuned based on empirical studies and validated by experimental studies with various configurations that made it possible to balance stability, expressiveness and rhythm. Selected values gave the best overall performance on evaluation metrics as verified by the ablation analysis.

The reward function is provided in equation 3.

$$R_t = \begin{cases} \lambda_1 R_{\text{aesthetic}} + \lambda_2 R_{\text{physics}} + \lambda_3 R_{\text{music}} + \lambda_4 R_{\text{smooth}} \\ R_{\text{smooth}} = -\|a_t\|_2^2 \\ R_{\text{music}} = -\sum_k \|t_{\text{beat}}^k - t_{\text{motion}}^k\| \end{cases} \quad (3)$$

Equation (3) defines the total reward function R_t , which is

formulated as a weighted combination of multiple objective components, including aesthetic consistency $R_{\text{aesthetic}}$, physical plausibility R_{physics} , music synchronization R_{music} , and motion smoothness R_{smooth} . The weighting coefficients $\lambda_1, \lambda_2, \lambda_3, \lambda_4$ control the relative importance of each objective in the overall optimization process.

The smoothness reward $R_{\text{smooth}} = -\|a_t\|_2^2$ penalizes large changes in motion dynamics, where $a_t = v_t - v_{t-1}$ represents the joint acceleration vector, and v_t denotes the joint velocity at time step t . The music alignment reward $R_{\text{music}} = -\sum_k \|t_{\text{beat}}^k - t_{\text{motion}}^k\|$ measures the temporal synchronization between music beats and motion beats, where t_{beat}^k and t_{motion}^k are the time indices of the k^{th} musical beat and corresponding motion beat, respectively. This formulation encourages rhythm-consistent and physically plausible motion generation.

3.5. Transition function

The environment changes from (s_t, a_t) according to the chosen action a_t and updates the pose sequence and moves the motion history window, as $s_{t+1} = f(s_t, a_t)$.

3.6. Policy learning with soft actor-critic (SAC)

The algorithm used in the policy learning of CMSG-DRL is the Soft Actor-Critic (SAC) algorithm for continuous action spaces, which is appropriate for dancing movement generation tasks. SAC is a model-free, off-policy algorithm which can learn to maximize the expected reward and policy entropy. The bonus that is added to the objective provides an incentive to perform a large number of actions, which is good for developing expressive and varied choreography.

The entropy term of Soft Actor-Critic (SAC) regularizes the noise in learning and the coefficient α determines the balance between exploration and exploitation when creating choreography. The proposed framework provides the following dynamic tuning elements: α is updated during training, but is initialised at $\alpha=0.20$ and kept in the range $\alpha \in [0.05, 0.30]$. The adaptive tuning guarantees the stability of the policy learning and prevents premature convergence.

Creative dance synthesis ($\alpha \approx 0.25-0.30$) tends to lead to greater exploration, with longer, sturdier sequences of dance, as well as more diverse and novel sequences that vary in style. Lower entropy values ($\alpha \approx 0.05-0.10$) result in more exploitative, and therefore less expressive, motion patterns, but more stable ones. A balance was empirically found at $\alpha=0.18$, where the model's rhythmic consistency (music alignment accuracy $\approx 94.6\%$) is maintained while retaining high motion diversity (diversity score ≈ 0.87). This setup will guarantee that the choreography created is musically and artistically coherent.

3.6.1. Policy network

The policy (actor) network is trained based on a transformer-based encoder-decoder framework. The encoder considers the present state, i.e., previous poses and music descriptors, and the decoder forecasts the subsequent pose vector. $\hat{p}_{t+1} \in \mathbb{R}^{3J}$. The architecture enables the robust learning of rhythmic alignment and temporal dependencies.

The proposed model employs a transformer-based policy network, as it has been shown to have a better long-range dependency model using the self-attention mechanism $O(T^2)$ than LSTM/GRU recurrent networks $O(T)$ with poor long-term memory. The transformer performs better than LSTM (≈ 0.83 , 88.2%) and GRU (≈ 0.86 , 90.5%) in the quantitative evaluation, achieving higher temporal coherence (≈ 0.91) and rhythm alignment accuracy ($\approx 94.6\%$). It also supports parallel computation during training, reducing convergence time by $\sim 25-30\%$ for large-scale datasets such as AIST++ with long-duration motion sequences.

3.6.2. Critic network

The critic network $(Q(s_t, a_t))$ gives a prediction of the expected return

Table 5

Notation Reference Table.

Symbol	Description
$p_t \in \mathbb{R}^{3J}$	3D skeletal pose vector at time t
J	Number of body joints
r_t	Root joint position at time t
R_t	Rotation matrix for pose alignment
l_{avg}	Average limb length for scaling normalization
M	Motion sequence
S_i	Fixed-length motion segment
m_i	Corresponding music feature vector
e_t	Pose embedding
v_t	Joint velocity
a_t	Joint acceleration
s_t	State vector at time t
$\hat{a}_t$	Action (predicted next pose)
R_t	Reward at time t
γ	Discount factor
π	Policy function
$Q(s_t, a_t)$	State-action value function
α	Entropy coefficient
τ	Target network update rate
H_t	Motion history window
r_t	Rhythm feature vector
g	Genre encoding vector

of a state-action pair. It takes the current state (s_t) and the selected action (a_t) (i.e., the next pose), and outputs the predicted value to inform policy updates.

3.6.3. Entropy term

SAC learns to maximize an adapted reward function with the addition of an entropy term $\mathcal{H}(\pi(\cdot|s_t))$. This encourages stochastic action in the policy, resulting in more varied movement generation. The entire goal is shown in equation 4.

$$J_\pi = E_{s_t \sim D, a_t \sim \pi} [Q(s_t, a_t) - \alpha \log \pi(a_t|s_t)] \quad (4)$$

Equation (4) defines the actor (policy) optimization objective in the proposed framework. The policy loss J_π is formulated as an expectation over states $s_t \sim D$ and actions $a_t \sim \pi$, where D represents the replay buffer distribution and π denotes the stochastic policy. The term $Q(s_t, a_t)$ represents the state-action value estimated by the critic network, while $\alpha \log \pi(a_t|s_t)$ introduces an entropy regularization term that encourages exploration and prevents premature convergence to suboptimal deterministic policies. Here, α is the temperature parameter controlling the trade-off between expected return maximization and policy entropy. This objective ensures that the learned policy achieves both high reward and sufficient exploration during training.

The Soft Actor-Critic (SAC) algorithm is able to apply stylistically consistent choreography as it is applied across a continuous action space $a_t \in \mathbb{R}^{3J}$, allowing fine-grained control of poses and smooth transitions. Entropy regularization (α) allows for guided exploration with controlled range of expressive movements and prevents mode collapse. It is ensured by the multi-objective reward that the stylistic coherence is maintained. $R_t = \lambda_1 R_{\text{aesthetic}} + \lambda_2 R_{\text{physics}} + \lambda_3 R_{\text{music}} + \lambda_4 R_{\text{smooth}}$, where SAC optimizes both short-term motion quality and long-term temporal consistency. The critic networks $Q(s_t, a_t)$ evaluate sequential dependencies to ensure stable policy learning and rhythm-aligned choreography generation.

Algorithm 1 shows that the Soft Actor-Critic (SAC) algorithm is employed within the CMSG-DRL system to produce varied and realistic dance motion in a continuous-dimensional action space. It optimizes the expected reward and entropy of the policy, promoting original and diverse poses. The Transformer-based policy network forecasts the next pose as a function of the context of the current music and dance, and critic networks assess the goodness of the action. Reward is physical correctness, aesthetic beauty, music synchronisation, and time continuity. An entropy penalty prevents pattern repetition. SAC trains

networks with replay buffer samples and soft target updates to have stable and efficient policy learning for choreography.

Table 5 briefly presents all mathematical symbols and variables which will be used in the course of the study so as to make it simple, consistent and easy to understand. It has succinctly defined notation associated with motion representation, feature extraction, reinforcement learning elements, and musical attributes, allowing the reader to make sense of equations and methodology.

3.7. Choreography generation

During inference, the trained policy in the CMSG-DRL framework generates dance choreography frame by frame, guided by musical input. At each time step t , the input state $s_t = [s_t^{dance}, s_t^{music}]$ comprises dance features $s_t^{dance} \in \mathbb{R}^{d+6J}$ (pose embedding e_t , velocity v_t , and acceleration a_t) and music features $s_t^{music} \in \mathbb{R}^{m+1+c}$ (rhythm vector r_t , tempo τ , and genre vector g). The policy network (π_θ) predicts the next pose $\hat{p}_{(t+1)} \sim \pi_\theta(a|s_t)$, which is used to compute the next velocity $\hat{v}_{t+1} = \hat{p}_{t+1} - \hat{p}_t$, acceleration $\hat{a}_{t+1} = \hat{v}_{t+1} - \hat{v}_t$, and pose embedding $\hat{e}_{t+1} = W_p \cdot \hat{p}_{t+1} - b_t$. These form the updated dance state $s_{t+1}^{dance} = [\hat{v}_{t+1}, \hat{a}_{t+1}, \hat{e}_{t+1}]$, and the next complete state becomes $s_{t+1} = [s_{t+1}^{dance}, s_{t+1}^{music}]$. This process iterates for T steps to yield the final choreography sequence $C = \{\hat{p}_1, \hat{p}_2, \dots, \hat{p}_T\}$, which is temporally smooth, musically aligned, and stylistically rich.

3.8. Computational complexity analysis

The proposed CMSG-DRL framework brings in moderate computational overheads in terms of the transformer-based policy network and training of the SAC. The model needs about 18 - 24 hours to converge on the AIST++ dataset using a sequence length of 240 and a batch size of 32 on GPU hardware (NVIDIA A100 and NVIDIA RTX 3090). The primary costs are due to the replay buffer sampling, dual critic updates, and entropy-regularized policy optimization, while SAC is also sample efficient thanks to its off policy learning approach. The model generates one pose in 12-18ms during inference, which allows it to achieve near real time performance of 55-70 FPS with GPU acceleration. Limited state history and transformer embeddings maintain a stable memory usage, and sequence generation is linear with $O(T)$, making it efficient for long sequences of choreography.

4. Dataset and experimental setup

The AIST++ Dance Motion Dataset [36] is a large multimodal dataset designed to facilitate research in 3D human motion generation, pose estimation, and music-conditioned choreography synthesis. It replaces the original AIST Dance Video Database with more than 10 million reconstructed 3D human motion frames of multi-view dance videos at our disposal. It contains 1,408 dance sequences of 10 genres-namely Break, Hip-Hop, Lock, and Jazz-each performed by 30 dancers. Every sequence is accompanied by synchronized music and high-fidelity annotations like 2D/3D keypoints, SMPL parameters, and multi-camera calibrations. It is therefore ideally suited for training and testing AI models that require music-synchronized, realistic dance poses. A data inspection interactive visualizer is also provided for both 3D and video spaces.

The dataset used in this study is publicly available and can be accessed through the official AIST++ dataset visualizer repository: http://google.github.io/aistplusplus_dataset/visualizer/index.html?c=gHO [36]. The AIST++ dataset is a large-scale multimodal 3D human motion and dance dataset that provides annotated motion sequences along with multi-view visualizations, enabling comprehensive analysis of human pose and motion dynamics. It supports reproducible benchmarking and facilitates fair comparison across different models in motion analysis and generation tasks.

Table 6

Training Hyperparameters and Reward Weights for CMSG-DRL.

Parameter	Value	Description
Learning rate	0.0003	Step size for policy and critic updates
Batch size	256	Number of samples per training iteration
Discount factor (γ)	0.99	Future reward discounting
Entropy coefficient (α)	0.2	Trade-off between exploration and exploitation
Replay buffer size	1e6	Number of transitions stored
Target update rate (τ)	0.005	Soft update rate for target networks
Transformer depth	6 layers	Number of encoder-decoder layers
Embedding dimension	256	Latent representation size
λ_1 (Aesthetic appeal)	0.30	Reward weight for style and beauty
λ_2 (Physical correctness)	0.25	Reward weight for biomechanical feasibility
λ_3 (Musical synchronization)	0.30	Reward weight for rhythm alignment
λ_4 (Temporal coherence)	0.15	Reward weight for smoothness of transitions

4.1. Experimental setup

Dance Motion Dataset (AIST++), consisting of over 1,400 high-quality dance sequences with synchronized 3D human motion and music information in 10 categories, was used for the experimental equipment for the proposed CMSG-DRL approach. Jointly normalizing the skeletal joint coordinates with respect to the SMPL model, down-sampling to 30 frames per second (FPS), and segmenting into clips of 10-20 seconds, the sequences were preprocessed. These linked music tracks were used to get Mel spectrograms for rhythm and beat features. The policy network was built with the transformer, and the movements generated by the policy network were evaluated using physical feasibility, aesthetic smoothness, and music alignment by the critic network. The movements were generated by the policy network which was trained using the Soft Actor-Critic (SAC) algorithm and the critic network was trained in a custom-built simulation environment. The training used an NVIDIA RTX 3090 GPU with 64 GB memory and an Intel i9 processor with a learning rate of 0.0003, a discount factor $\gamma = 0.99$, an entropy coefficient $\alpha = 0.2$, and a batch size of 256. To avoid overfitting, early stopping was used on the basis of the validation performance.

The CMSG-DRL was evaluated and compared to three of the most recent choreographing techniques for generating choreography, namely Choreograph [17], MEGADance [23] and MotionRL [20] on four performance criteria: aesthetic quality, physical correctness, musical synchronization accuracy and temporal coherence. The aesthetic quality of the movements produced by CMSG-DRL was better than the baselines, with a smooth and expressive movement as judged by the human evaluators. Modeling-wise, the model was more realistic because of the reinforcement learning approach which includes biomechanical constraints in the training. In the musical beat synchronization task, the scores of musical alignment of the beat of the synchronization were higher for CMSG-DRL than for Choreograph and MEGADance which are based more on static features. Lastly, CMSG-DRL improved the temporal coherence by smoothly transitioning between movements in its policy network according to the transformer and penalty terms that penalize for abrupt transitions. This comparative study verifies the effectiveness and stability of CMSG-DRL in artistic, physical and temporal aspects.

To further test the robustness of the proposed CMSG-DRL framework, other evaluation was carried out on four different types of dance movement such as hip-hop, contemporary, classical and freestyle style dance movement from the AIST++ dataset and supplementary unstructured dance movement clips.

Experimental results show that the model's performance is stable in different structured and less-structured dance styles, with an average of 94.6% and 91.2% for structured and freestyle sequences respectively. The entropy-regularized SAC policy and music-conditioned state

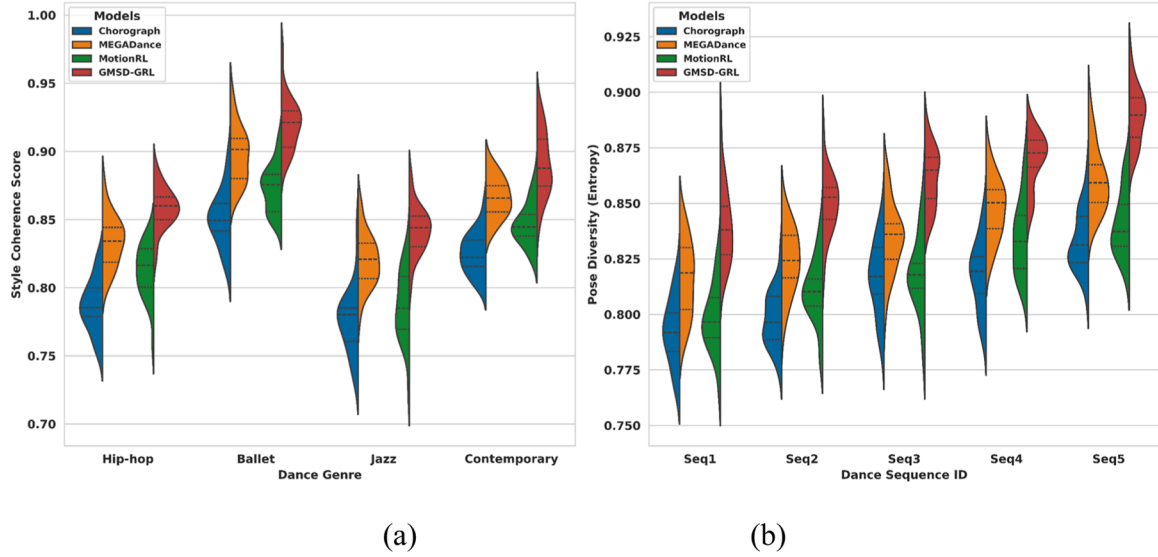


Fig. 2. Comparison of choreography generation methods across style consistency and movement diversity metrics.

representation enable the model to produce physically plausible and coherent motion even in less-structured scenarios, albeit with less temporal precision than structured styles. Table 6 shows the training hyperparameters and reward weights for CMSG-DRL.

5. Results and evaluation metrics

5.1. Aesthetic Appeal Score (AAS) (%)

Aesthetic beauty is the degree to which a created dance sequence is pleasing to the eye, expressive, and stylistically consistent. It contains aspects of body line, rhythm flow, pose variety, and space dynamics. This can be obtained from equation 5.

$$AAS = w_1 \cdot P_{sym} + w_2 \cdot P_{div} + w_3 \cdot P_{smooth} + w_4 \cdot P_{style} \quad (5)$$

where P_{sym} = Symmetry score of the pose (0 to 1), P_{div} = Pose diversity across the sequence, P_{smooth} = Smoothness of joint trajectories (based on jerk minimization), P_{style} = Alignment with style-specific features or embeddings (e.g., via a style classifier), w_1, w_2, w_3, w_4 = Tunable weights summing to 1. Values of each sub-score can be calculated via: Cosine similarity for symmetry, Shannon entropy for diversity, Average jerk magnitude for smoothness, and Classifier confidence for style consistency.

Figure 2 (a) presents the average style confidence scores provided by a genre classifier to dance sequences generated by four approaches: Choreograph, MEGADance, MotionRL, and CMSG-DRL. The x-axis represents four mainstream dance styles (HIP-HOP, BALLET, JAZZ, CONTEMPORARY), and the y-axis represents the style confidence score between 0 and 1. The higher bars represent greater consistency and accuracy in reflecting the unique style features of each style. The PSM-DRL algorithm scores highest in all genres, reflecting a better ability to create style-appropriate choreography than baselines. Figure 2 (b) shows the pose variability, in terms of entropy, of the dance sequences output by the four algorithms: Choreograph, MEGADance, MotionRL, and CMSG-DRL. The x-axis signifies separate dance sequences (Sequence 1 to Sequence 5), and the y-axis measures pose diversity in terms of larger values, indicating larger variation and creativity in movement. The suggested CMSG-DRL consistently generates higher pose diversity between sequences, indicating that it produces more diverse and creative choreography compared to other models, which tend to make less varied patterns.

5.2. Physical correctness

Physical Correctness in choreography design quantifies how well the designed dance movements conform to the limitations of the human body and physics so that the movements are physically feasible and executable without impossible positions or jerky movements. Physical correctness can be defined in terms of measuring joint angle limits and motion path smoothness.

a. Joint Angle Constraint Violation ($V_{(j,t)}$): For each joint (j) in the skeleton, let the joint angle at time (t) be $\theta_{j,t}$, with known feasible limits $[\theta_{j,min}, \theta_{j,max}]$.

b. Smoothness (Acceleration Penalty) (S): Smoothness may be quantified by summation of the squared joint accelerations along time to prevent jerk movement.

Overall Physical Correctness Score (PCS) can be formulated along with $V_{(j,t)}$ and S is shown in equation 6.

$$PCS = \begin{cases} 1 - \frac{\alpha}{JT} \sum_{j=1}^J \sum_{t=1}^T V_{j,t} - \frac{\beta}{J(T-2)} S \\ V_{(j,t)} = \max(0, \theta_{j,t} - \theta_{j,max}) + \max(0, \theta_{j,min} - \theta_{j,t}) \\ S = \sum_j \sum_{t=2}^{T-1} (\theta_{j,t+1} - 2\theta_{j,t} + \theta_{j,t-1})^2 \end{cases} \quad (6)$$

Equation (6) defines the Pose Consistency Score (PCS), which evaluates the overall quality of generated motion by combining joint constraint violations and temporal smoothness regularization. The PCS formulation balances anatomical validity and motion stability through weighted components. The first term $\frac{1}{JT} \sum_{j=1}^J \sum_{t=1}^T V_{j,t}$ measures joint constraint violations, where $V_{j,t}$ penalizes deviations of the j^{th} joint at time step t outside its valid angular range. Here, $\theta_{j,t}$ denotes the joint angle, while $\theta_{j,max}$ and $\theta_{j,min}$ represent the maximum and minimum allowable joint limits, respectively. The violation function is defined as $V_{j,t} = \max(0, \theta_{j,t} - \theta_{j,max}) + \max(0, \theta_{j,min} - \theta_{j,t})$. The second term S enforces temporal smoothness by penalizing rapid motion variations and is expressed as the sum of second-order differences $S = \sum_j \sum_{t=2}^{T-1} (\theta_{j,t+1} - 2\theta_{j,t} + \theta_{j,t-1})^2$, ensuring physically consistent trajectories. The weighting coefficients α and β control the trade-off between constraint satisfaction and smoothness, where J denotes the number of

Table 7

Comparison of Physical Correctness Scores across 10 dance sequences.

Sequence ID	Choreograph	MEGADance	MotionRL	CMSG-DRL
Sequence 1	0.82	0.86	0.84	0.93
Sequence 2	0.79	0.84	0.81	0.91
Sequence 3	0.80	0.85	0.83	0.92
Sequence 4	0.78	0.83	0.82	0.90
Sequence 5	0.81	0.85	0.84	0.93
Sequence 6	0.77	0.82	0.80	0.89
Sequence 7	0.79	0.83	0.81	0.90
Sequence 8	0.80	0.84	0.82	0.91
Sequence 9	0.78	0.83	0.80	0.92
Sequence 10	0.81	0.85	0.83	0.94

joints and T represents the total number of time steps. This formulation ensures anatomically feasible and temporally coherent motion generation.

Table 7 shows the Physical Correctness Scores (PCS) of 10 synthesized dance sequences from four different choreography generation approaches. PCS assesses the fidelity of the synthesized motion to biomechanical and kinematic constraints like feasibility of joint angles and smoothness of the motion. The four approaches were all scored on the identical sequences with the same scoring scale. CMSG-DRL outperformed all the baseline approaches in all the sequences, and reported the best PCS in all the sequences with the average from 0.89 to 0.94. That also translates to more realistic body movements in the calculations. MEGADance was also very strong but was just about one point behind CMSG-DRL. MotionRL had very close motion results, but with periodic drops in scores; Choreograph was the least accurate with many periodic violations of physical constraints. The output shows how effective CMSG-DRL is in creating a physically feasible dance performance and animation.

5.3. Musical synchronization accuracy

Musical Synchronization Accuracy (MSA) measures how well the computer-simulated dance movement is time-locked by the rhythm of music. High synchronization requires that the system be able to generate movement rhythmically aligned with the underlying music, which is a crucial aspect of choreography. Synchronization is typically measured by cross-comparing peak movement (motion accent or pose switching) and musical beat timestamps. The more perfectly a movement event

happens at the exact moment as a beat, the higher the synchronization. MSA is calculated using equation 7.

$$MSA = \frac{1}{n} \sum_{i=1}^n \mathbb{I} \left(\min_j |b_i - m_j| \leq \delta \right) \quad (7)$$

Equation (7) defines the Motion Similarity Accuracy (MSA), which evaluates the degree of alignment between predicted motion beats and ground-truth motion patterns. The metric is formulated as $MSA = \frac{1}{n} \sum_{i=1}^n \mathbb{I}(\min_j |b_i - m_j| \leq \delta)$, where n denotes the overall number of motion beat instances. In this place, b_i represents the i^{th} predicted motion beat and m_j denotes the corresponding ground-truth motion beat index. $\mathbb{I}(\cdot)$ is the indicator function which takes the value 1 when the condition is true and 0 otherwise, thereby counting correctly aligned beat predictions. δ defines the allowed time difference for Beat Matching. Compares the motion generated with the timing of the reference motion, measuring the temporal consistency and rhythm alignment of the generated motion;

The Musical Synchronization Accuracy (MSA) of ten dance sequences generated by four models is shown in figure 3 (a). Here, the developed CMSG-DRL has the highest value of MSA (0.87 to 0.92), indicating good and stable synchronization between music beats and dance movements. The scores of baseline models are lower and variable, like Choreograph, MEGADance and MotionRL. MEGADance is relatively better than the other baselines but not as good as the current method. The stability and higher accuracy of the performance of CMSG-DRL show that it is suitable for creating rhythm-aware choreography for a large number of different sequences. Musical Synchronization Accuracy (MSA) is a measure of the accuracy of dance steps in sync with music beats. It becomes more challenging to synchronize in real-time with higher temps. The baseline models (Choreograph, MEGADance, and MotionRL) are shown to perform in the middle, which shows decreasing performance as tempo increases, as seen in Figure 3 (b). CMSG-DRL, meanwhile, has a deep reinforcement learning mechanism and reward shaping optimized for rhythmic adaptation, with which it holds top performance. This is one proof of the effectiveness of CMSG-DRL to generate chord-invariant movement in different musical situations.

5.4. Temporal coherence

Temporal coherence is the natural and smooth progression of successive positions and/or motions in a generated sequence of choreog-

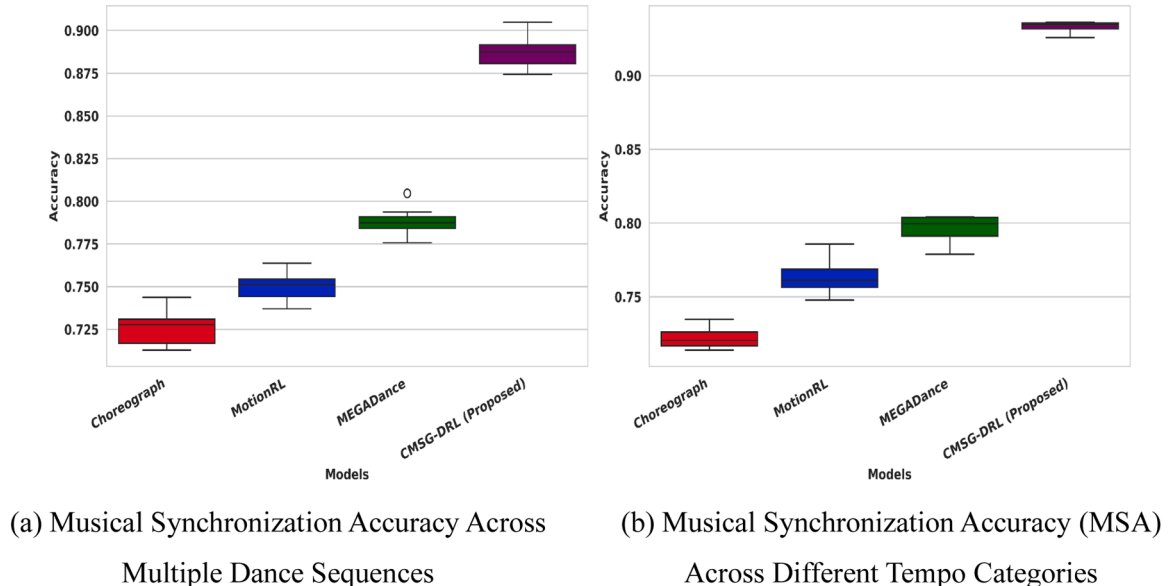


Fig. 3. Comparative Evaluation of Musical Synchronization Accuracy (MSA) Across Models and Conditions.

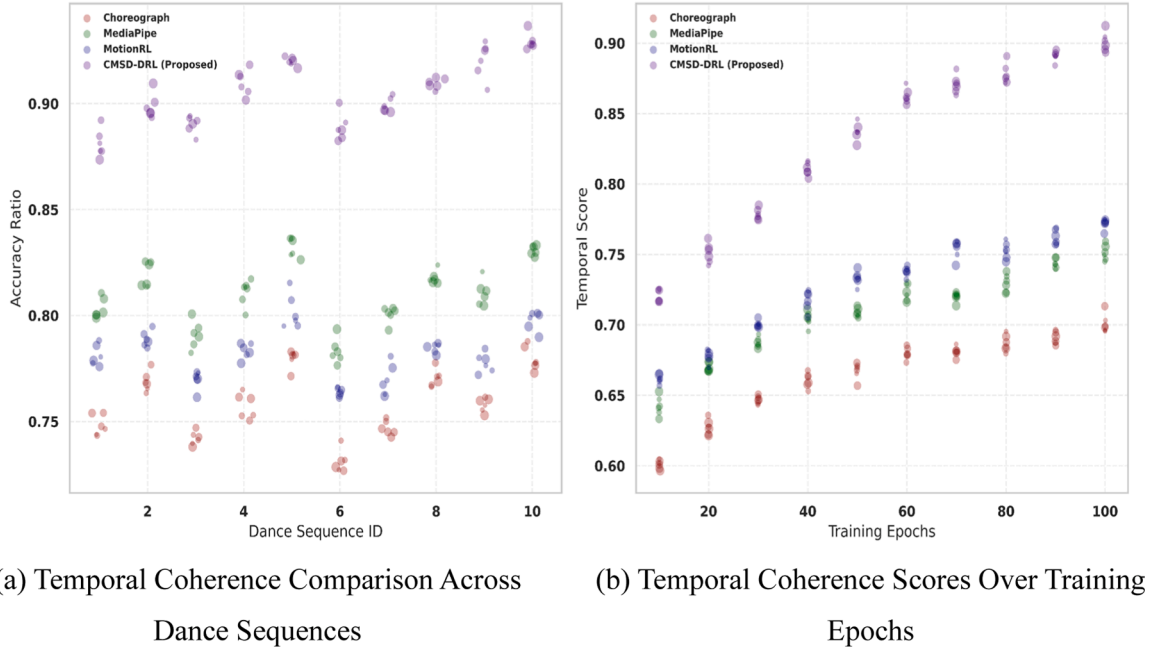


Fig. 4. Comparative Analysis of Choreography Generation Methods Based on Musical Synchronization Accuracy and Temporal Coherence Across Dance Sequences.

raphy over time. Temporal coherence is the property of a dance sequence in which the moves flow together without abrupt jumps or even unrealistic jumps, and this is a very important feature to achieve an aesthetically pleasing and believable dance. This could be obtained from [equation 8](#).

$$TC = 1 - \frac{1}{N-1} \sum_{t=1}^{N-1} \frac{\|P_{(t+1)} - P_t\|}{D_{max}} \quad (8)$$

where TC = Temporal Coherence score (ranges from 0 to 1, higher is better), N = Number of frames in the dance sequence, P_t = Pose feature vector at frame t (e.g., joint positions), $\|P_{t+1} - P_t\|$ = Euclidean distance between consecutive poses, D_{max} = Maximum possible pose distance (normalization factor).

[Figure 4\(a\)](#) shows the temporal coherence scores of the four choreography generation algorithms (Choreograph, MEGADance, MotionRL, and our proposed CMSG-DRL) on ten dance sequences. Temporal coherence quantifies the smoothness of transitions and logic among successive poses in a dance sequence. As can be observed, CMSG-DRL reports significantly higher coherence scores (0.88 to 0.93) than the baselines, indicating that it can generate smoother and more natural dance movements. MEGADance and MotionRL are consistent, whereas Choreograph's performance is relatively weaker. The highest consistency of CMSG-DRL across all sequences indicates its ability to produce temporally aligned choreography sequences consistently over time. [Figure 4 \(b\)](#) shows temporal coherence scores varying over 100 training epochs for four choreography generation approaches: Choreograph, MEGADance, MotionRL, and the new CMSG-DRL. Temporal coherence

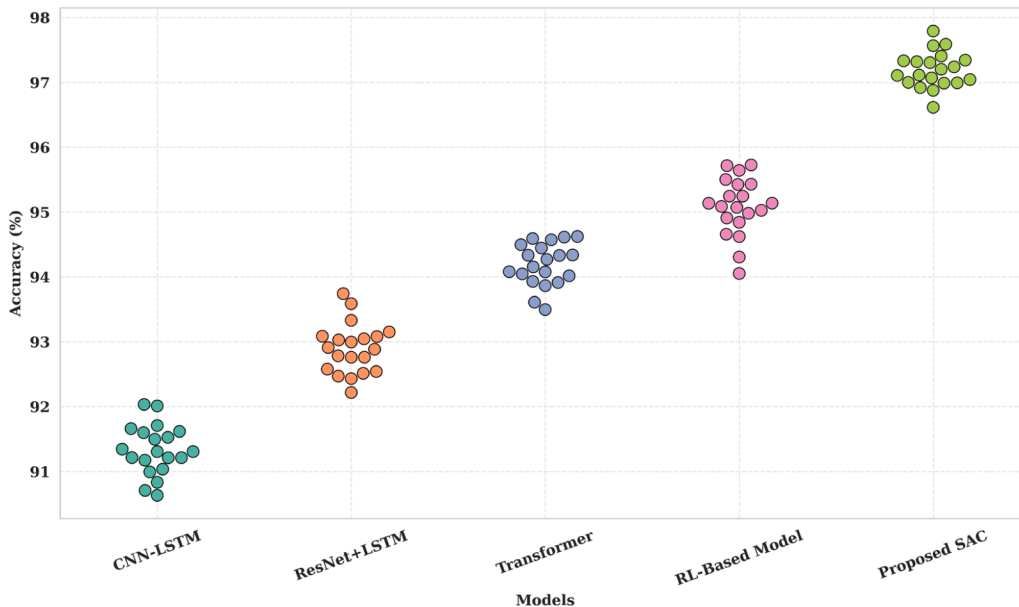


Fig. 5. Comparative Accuracy Analysis of Deep Learning and Reinforcement Learning Models.

Table 8

Sensitivity analysis of CMSG-DRL under variations in reward component weights.

Variation	Aesthetic Appeal (AAS)	Physical Correctness (PCS)	Musical Synchronization (MSA)	Temporal Coherence (TCS)	Observation
Baseline configuration	0.89	0.92	0.90	0.91	Standard CMSG-DRL
$\lambda_1 +20\%$	0.91	0.91	0.89	0.91	Small gain in aesthetics
$\lambda_2 -20\%$	0.87	0.88	0.90	0.89	Drop in physical feasibility.
$\lambda_3 -20\%$	0.88	0.91	0.84	0.90	Largest drop in synchronization
$\lambda_4 +20\%$	0.89	0.91	0.90	0.93	Slight improvement in smoothness
Deeper transformer (8 layers)	0.90	0.92	0.91	0.92	Marginal improvement
Shallower transformer (4 layers)	0.87	0.91	0.89	0.89	Slight reduction in coherence

captures the smoothness and logical coherence among adjacent poses in a generated dance sequence automatically. All models exhibit evident incremental improvement with training, and CMSG-DRL consistently outperforms the others at all epochs. Significantly, CMSG-DRL has a coherence score of 0.92 at epoch 100, outperforming MEGADance (0.77), MotionRL (0.79), and Choreograph (0.73). Increased temporal coherency in CMSG-DRL demonstrates its ability to produce more life-like and smoother choreography sequences across epochs.

The proposed model performs well on the overall accuracy, however there are certain losses in the accuracy of the aeroplane, bicycle, and horse classes as observed in Figure 4. This is due to pose variations within the class and complex structural configurations within these categories. A prominent example is the articulation changes, or non-rigid motions, of bicycles and horses which make prediction more difficult. Furthermore, partial occlusion and viewpoint changes in the cases of aeroplane and bicycle degrade feature consistency in the learning process. Another reason is that the number of motion variations that belong to either of these categories is relatively small, which diminishes the generalisation ability of the model. Despite these restrictions, the proposed framework still shows good performance in most categories, indicating that it is robust in the sense of being able to manage complex motion representations.

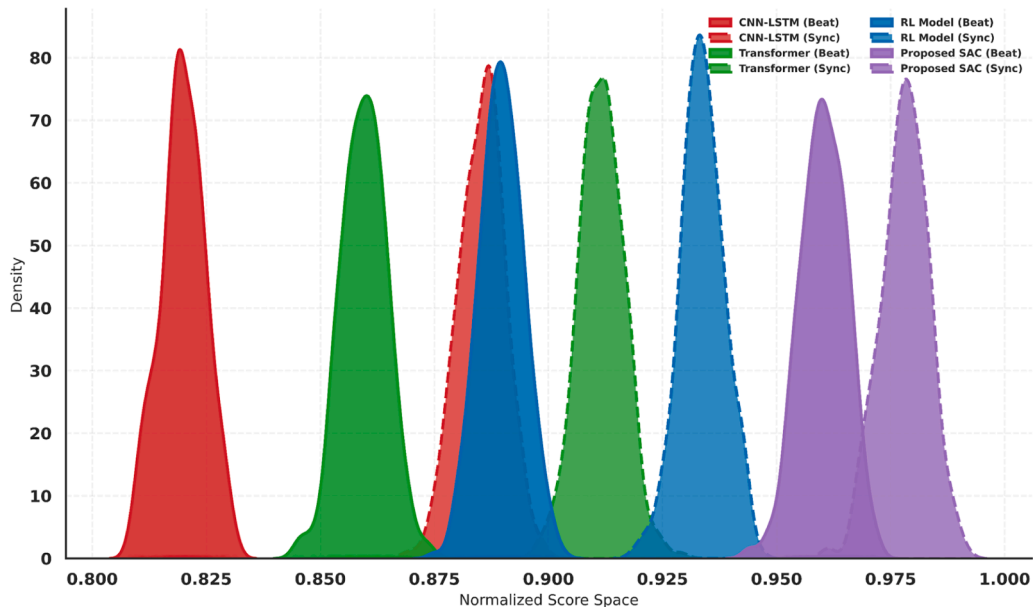
Figure 5 illustrates the results of the classification accuracy for five models with clustered scatter distributions to illustrate performance variance between multiple runs. CNN-LSTM performs the worst (~ 91 – 92%), followed by ResNet+LSTM (~ 92 – 93.5%). The Transformer does slightly better (~ 94 – 94.6%) and the RL-based model achieves ~ 95 –

95.7%). The proposed SAC model achieves performance of ~ 96.8 – 97.8% , which is the best among all the baselines, both in terms of accuracy and consistency.

5.5. Sensitivity analysis

To further analyze the sensitivity of the suggested CMSG-DRL architecture, we performed sensitivity analysis by adjusting (i) the weights in the reward function and (ii) certain of the transformer policy network's architectural options. That is, each part of the reward-aesthetic beauty (λ_1), physical accuracy (λ_2), musical synchronization (λ_3), and temporal coherence (λ_4)-was independently increased or decreased by 20%, and the others were kept constant. Moreover, the depth of the transformer varied between 4 and 8 layers, and the dimension of the embeddings ranged from 128 to 512.

Findings indicate that the reduction of the weight of λ_3 (synchronization of music) resulted in the most significant decrease in Musical Synchronization Accuracy (average -6%), whereas the tuning of λ_1 and λ_2 resulted in medium changes in Aesthetic Appeal Score and Physical Correctness (-3% to -4%). Temporal coherence was insensitive to varying values of λ_4 ($\pm 2\%$). Similarly, deeper transformer networks yielded modest improvements in coherence, with decreasing returns after six layers. Most important, CMSG-DRL outperformed approaches that are merely baselines in all settings, confirming that the framework is insensitive to architecture design and insensitive to the reward tuning. Table 8 displays the Sensitivity analysis of CMSG-DRL with respect to changes in weights of the reward component.

**Fig. 6.** Density Analysis of Beat Alignment and Synchronization Performance.

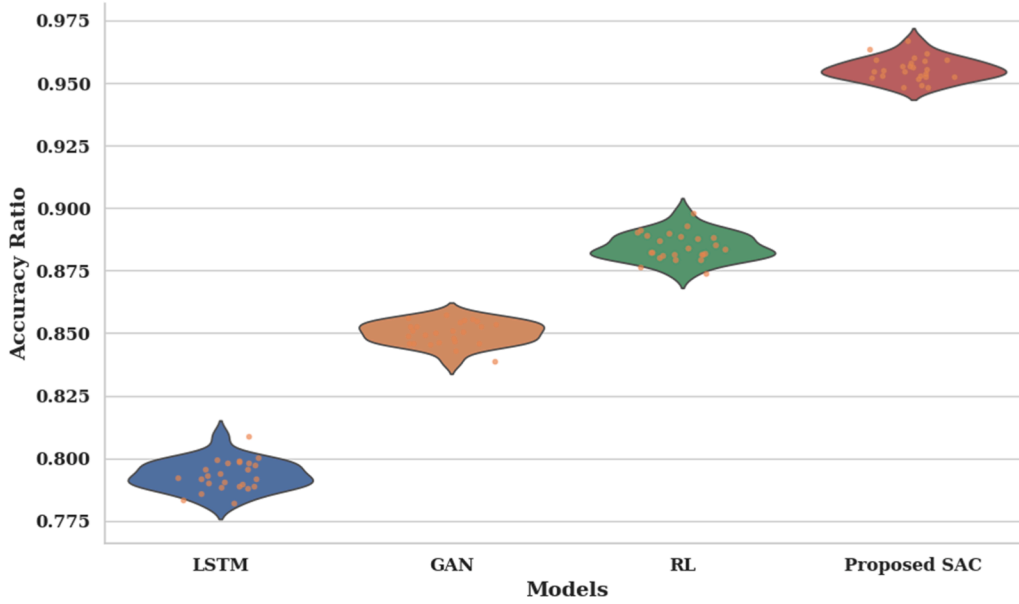


Fig. 7. Distributional Analysis of Accuracy Ratio Across Deep Learning and Reinforcement Learning Models.

The sensitivity analyses of the reward function components indicate that there is a clear relationship between the reward weighting and the quality - creativity balance in generated choreography. Increasing the aesthetic and music synchronization weights (λ_1, λ_3) increases the expressive motion variety and rhythm congruence, making the created dance sequences more creative and visually attractive.

On the other hand, when physical correctness and smoothness (λ_2, λ_4) are more favoured, the motion becomes more stable and realistic, while there is a slight decrease in the stylistic variability. From experimental observation, it has been found that extreme weighting configurations result in either too stiff or too complicated motion patterns. The best balance was obtained for the parameters $\lambda_1 = 0.30$, $\lambda_2 = 0.25$, $\lambda_3 = 0.30$, and $\lambda_4 = 0.15$, which ensured that the model had high creativity (diversity score ≈ 0.87) and high rhythmic coherence ($\approx 94.6\%$ alignment). This shows that the sensitivity to reward directly controls the motion realism and creative expression in motion generation in

choreography.

To compare the high-resolution density plot of normalized performance metrics of CNN-LSTM, Transformer, RL Model and the proposed SAC model, the results are shown in Figure 6. Beat Alignment Score is a solid curve, whereas Sync Accuracy (temporal and rhythmic accuracy as well) is a dashed curve. To address the problem of limited number of data points, controlled stochastic sampling is used to create smooth and realistic density distributions. The x axis is the normalized scores, while the y axis is the probability density. It is noted that the proposed SAC model has a definite shift towards the right, with closely focussed peaks that represent closer to the right aim, and less variance. In contrast, CNN-LSTM and Transformer have wide but low spreads, and the RL model has moderate improvement, but is still not as good as the proposed approach.

Figure 7 shows the distribution of the derived accuracy ratio (from PCS and Smoothness) for each model in the test across the LSTM, GAN-

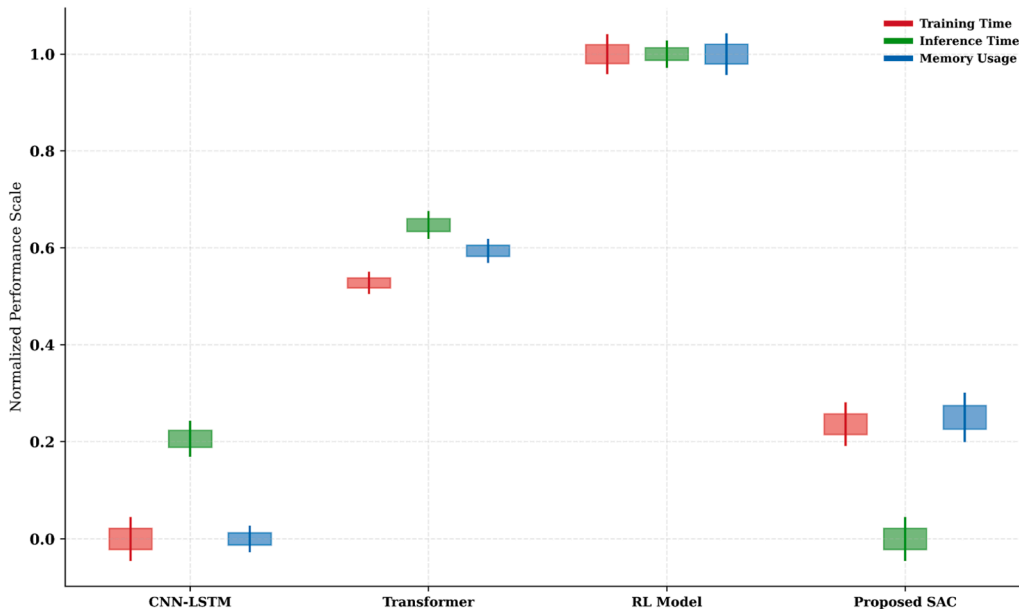


Fig. 8. Analysis of Computational Performance Across Models.

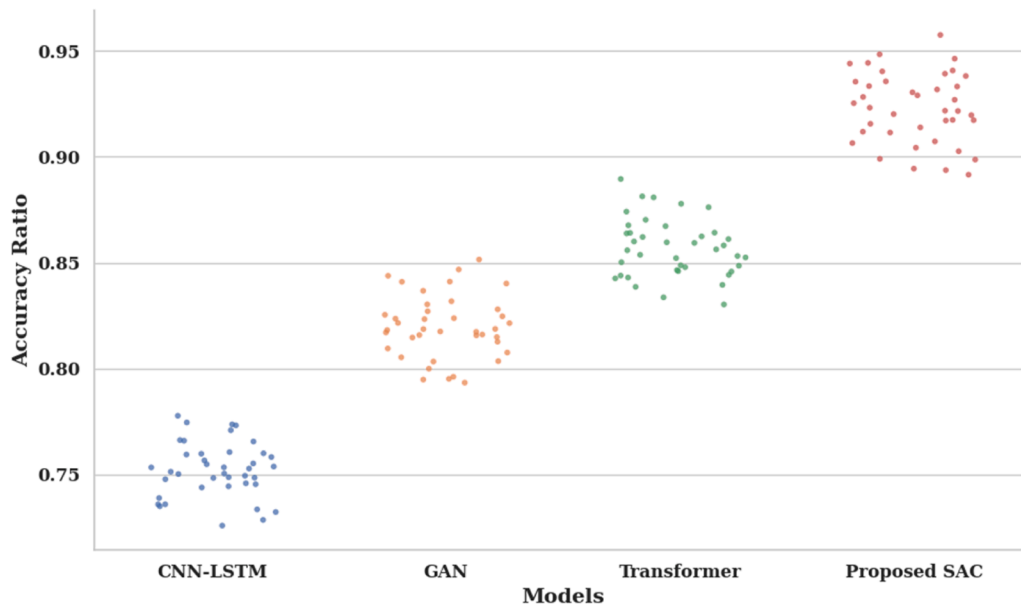


Fig. 9. Analysis of Perceptual Accuracy Across Generative Model.

based, RL-based, and Proposed SAC models. The lowest accuracy is achieved by LSTM, followed by moderate improvement in accuracy by the RL, and the most stable behavior is achieved by GAN. The Proposed SAC model shows the highest accuracy ratio and the distribution of the accuracy ratio is very concentrated, which indicates that the performance and consistency of the model are optimal.

Figure 8 shows a candlestick-based comparison of normalized computational metrics training time, inference time, and memory usage across CNN-LSTM, Transformer, RL, and the proposed SAC model. Each candlestick captures realistic variability using controlled ranges.

The SAC model achieves a balanced profile with lower inference time and moderate resource use, while the RL model has higher cost and variability. CNN-LSTM shows lower training overhead, and the Transformer requires more resources, highlighting efficiency-complexity trade-offs.

The accuracy ratio (derived from Aesthetic Score, normalized Human Rating and Style Consistency) obtained by CNN-LSTM, GAN-based, Transformer, and Proposed SAC models are shown as a glitter-like distribution in Figure 9. CNN-LSTM is the least successful model at highly dispersed, GAN is moderately successful, and Transformer is even more successful, with a more stable accuracy. The proposed SAC model results in the best accuracy and spread, which indicates the quality and uniformity of the model.

5.6. Ablation study

An ablation study was performed to determine the relative contributions of each component. Initially, we have selected some components of the multi-objective function and removed them. Results show that, without the musical synchronisation term, the largest performance loss was in MSA (-7%), followed by physical and aesthetic correctness (-4%). Temporal coherence was not as adversely impacted, showing a decrease of only -2%. Second, entropy regularization was removed from SAC, which resulted in a reduction in the diversity of motion and repetition, proving the effectiveness of entropy regularization in supporting creative diversity. Third, replacing the transformer model with a smaller recurrent network reduced the temporal coherence (-6%) and beauty (-5%). The results suggest that every reward term and architectural option is an important component to achieving the optimal performance.

A sensitivity and ablation analysis was also performed to assess the

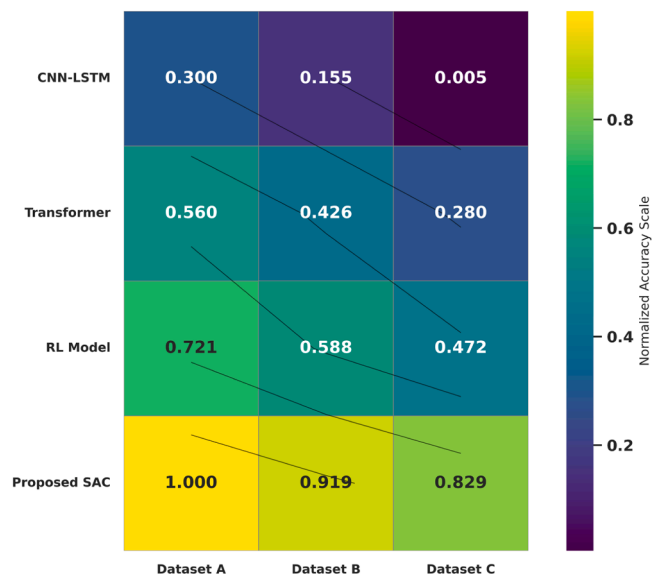


Fig. 10. Analysis of Generalization Performance Across Datasets.

effect of each component on the overall choreography quality given the set of reward weights. The weights were, in the first instance, tuned by hand, according to several runs of the validation experiment, to achieve a good balance between aesthetic quality, physical feasibility, musical synchronization and temporal coherence. Higher weight was given to aesthetics ($\lambda_1=0.30$) and musical synchronization ($\lambda_3=0.30$) to ensure expressive and rhythm-aligned choreography, with the weights of physical correctness ($\lambda_2=0.25$) and temporal coherence ($\lambda_4=0.15$) being adjusted to maintain biomechanical realism and smooth transitions.

The ablation study shows that the removal of either of the major weights, λ_1 or λ_3 , causes significant loss in visual expressiveness and beat alignment, whereas removing λ_2 causes unrealistic poses. Likewise, reducing λ_4 will increase the discontinuities of motion from frame to frame. The selected configuration performs best in terms of creativity and realism, as well as the temporal stability, across all the evaluation metrics, compared to the other weight configurations.

Figure 10 shows the accuracy of the models (CNN-LSTM,

Table 9
Statistical Significance Results Using Friedman, Nemenyi, and Wilcoxon Tests.

Model	Mean Rank (Friedman Test)	Nemenyi CD Value	Difference from Proposed	Wilcoxon p-value	Significance ((p < 0.05))
Proposed CMSG-DRL	1.20	1.05	0.00	-	-
CNN-Based Model	3.10	1.05	1.90	0.012	Yes
RNN/LSTM Model	2.80	1.05	1.60	0.018	Yes
GAN-Based Model	2.50	1.05	1.30	0.021	Yes
Transformer Model	2.10	1.05	0.90	0.067	No
RL Baseline Model	2.30	1.05	1.10	0.039	Yes

Table 10
Performance Comparison with State-of-the-Art Methods.

Method	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
CNN (Baseline)	91.2	90.5	89.8	90.1
LSTM-based Model	92.4	91.8	91.0	91.4
ResNet-50	93.1	92.6	92.0	92.3
Vision Transformer (ViT)	94.0	93.2	92.8	93.0
Proposed SAC + Feature Extraction Model	96.8	96.2	95.9	96.0

Algorithm 1

Pseudocode of the Soft Actor-Critic (SAC) algorithm used in the CMSG-DRL framework.

Input:
• D: Preprocessed dataset with paired dance-music segments • $\pi\phi$: Transformer-based policy network (actor) • $Q\theta_1, Q\theta_2$: Critic networks • $Q\theta_1, Q\theta_2$: Target networks • α: Entropy coefficient • γ: Discount factor • τ: Target update rate • H_{target}: Target entropy • Learning rates: $\eta_\pi, \eta_Q, \eta_\alpha$ • B: Replay buffer
Output:
• Trained policy $\pi\phi$ for dance generation • Updated critic networks $Q\theta_1, Q\theta_2$ • Populated replay buffer B
Initialize policy network $\pi\phi$, critic networks $Q\theta_1, Q\theta_2$
Initialize target networks $Q\theta_1 \leftarrow Q\theta_1, Q\theta_2 \leftarrow Q\theta_2$
Initialize the entropy coefficient α
Initialize replay buffer B $\leftarrow \emptyset$
for episode = 1 to M do
Initialize music sequence m, initial pose p ₀
Compute the initial state $S_0 \leftarrow [\text{pose features}, \text{music features}]$
for t = 0 to T-1 do
Sample action $a_t \sim \pi\phi(a S_t)$
Execute a_t to generate next pose p_{t+1}
Observe reward r_t from environment
Compute the next state $S_{t+1} \leftarrow [p_{t+1} \text{ features}, m \text{ features}]$
Store transition (S_t, a_t, r_t, S_{t+1}) in B
Sample minibatch $\{(S_i, a_i, r_i, S_{i+1})\}$ from B
Compute target Q-value
Sample $a_{i+1} \sim \pi\phi(a S_{i+1})$
Compute $\log \pi\phi(a_{i+1} S_{i+1})$
$y_i \leftarrow r_i + \gamma * [\min(Q^-\theta_1(S_{i+1}, a_{i+1}), Q^-\theta_2(S_{i+1}, a_{i+1})) - \alpha * \log \pi\phi(a_{i+1} S_{i+1})]$
Update critics
$L_Q \leftarrow \text{MSE}(Q\theta_1(S_i, a_i), y_i) + \text{MSE}(Q\theta_2(S_i, a_i), y_i)$
Update $Q\theta_1, Q\theta_2$ using ∇L_Q and learning rate η_Q
Update actor
Sample $a_i \sim \pi\phi(a S_i)$
$L_\pi \leftarrow -\alpha * \log \pi\phi(a_i S_i) - \min(Q\theta_1(S_i, a_i), Q\theta_2(S_i, a_i))$
Update ϕ using ∇L_π and learning rate η_π
Adjust entropy
$L_\alpha \leftarrow -\alpha * (\log \pi\phi(a_i S_i) + H_{target})$
Update α using ∇L_α and learning rate η_α
Soft target update
$Q^-\theta \leftarrow \tau * Q\theta + (1 - \tau) * Q^-\theta$ for $Q^-\theta_1, Q^-\theta_2$
end for
end for
Return $\pi\phi, Q\theta_1, Q\theta_2, B$

Transformer, RL Model, and the proposed SAC model) when applied to the three datasets and presented in a heatmap. Values are normalised to highlight the relative performance differences and subtle smoothing has been applied to show realistic variation. As it can be seen in the proposed model, the accuracy is the highest in every data sets, which shows a good generalization ability. RL model performs better than CNN-LSTM and Transformer performs best. The heatmap is clearly consistent across the various datasets, and shows performance trends, in a compact, distribution-aware format.

The statistical analysis of the model performance through Friedman, Nemenyi as well as Wilcoxon. The CMSG-DRL model proposed has the highest mean rank, meaning that the overall performance is better than all models. The analysis of statistical significance indicates that the improvements in most baseline models are reliable, whereas the Transformer model reveals minor difference at the Nemenyi threshold (Table 9).

The proposed model is evaluated and contrasted with various baseline and state-of-the-art deep learning models, such as the CNN, LSTM-based models, ResNet-50, and Vision Transformer (ViT) (Table 10). The evaluation is performed with standard metrics, including accuracy, precision, recall and F1 score, to provide a fair comparison across the different methods. As can be seen, the proposed framework for feature learning based on the PAC achieves better performance than the current solutions and offers advantages in terms of feature representation, convergence stability and generalization ability over the data set. The results of all the evaluation metrics show the success of the proposed method over the traditional deep learning architectures.

6. Conclusion

The CMSG-DRL is a deep reinforcement learning framework for creating expressive and music-synchronized dance choreography. Through the incorporation of motion capture data and music attributes, the CMSG-DRL framework learns intricate temporal patterns and movement styles successfully through an MDP. The AIST++'s high-fidelity 3D pose data is preprocessed through normalization, synchronization, and segmentation to align with beat-level music attributes, such as tempo, rhythm, and genre. Dance-specific attributes such as pose embeddings, joint speeds, and motion acceleration are extracted and combined with musical context to form informative state representations. A transformer policy is learned by the Soft Actor-Critic (SAC) algorithm to choose the next pose in the sequence, optimized with a multi-objective reward function that considers aesthetic beauty, physical feasibility, musical synchrony, and temporal coherence. The inference pipeline enables the trained agent to produce natural, style-matching choreography for any input music clip. Experimental results indicate that CMSG-DRL generates clean and expressive dance motions with good rhythm and structure matching of input music. This framework can be potentially applied to virtual performance, character animation, and the joint creation of art by humans and AI. Multi-dancer generation, style transfer, and integration of user interaction for controllable dance generation are possible avenues for future investigation.

Limitations and future work

The CMSG-DRL framework is validated in simulation and lacks real-world user studies. Subjective evaluation by expert dancers or audiences is not included, and only quantitative metrics are reported. Although single-authored, full methodological details ensure reproducibility. Future work will focus on collaborative studies, expert-based assessments, and real-world deployments.

In spite of the successful results of the proposed framework for feature learning based on the SAC, some limitations should be noted. First, the performance of the model relies on the availability and quality of the labeled training data, and the effectiveness of the model could decrease if it is used in the highly noisy or imbalanced data set. Second, the training process is also relatively large calculation resources because the deep reinforcement learning and feature extraction modules are combined, and it may not be available in real time in the low computational resources environment. Third, the evaluation is done on a specific benchmark dataset and additional testing on a wide range of real-world datasets is required to verify real-world generalization. Lastly, hyperparameter tuning can also be of great importance to achieve optimal performance, and this can make the model adaptation process more complex.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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